



Cross-Architecture Ensembles for Legal Information Processing

One method per task, the right inductive bias for each.

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TASK 4 WINNER · 1ST OF 33 RUNS

Task 4
Statute Entailment

96.3%

Accuracy on 79 of 82 questions. Our two leading runs, DU1 and DU2, share first place, and the best rival run reached 95.1%.

Validated at 93.0% on held-out questions before the unseen test set.

9
open-weight experts

3
model families

82
test questions

How the win was built

Nine experts from three families, each paired with a prompting strategy, vote on every question. The gain comes from disagreement between architectures, not from scale.

DeepSeek

R1 (671B)

A reasoning model that works through the question before it answers.

Llama

4 Maverick (400B) 4 Scout (109B)

3.3 (70B)

Three sizes, from a dense 70B to a large mixture of experts.

Qwen

2.5 (72B) 3 (235B)

Two generations of the Qwen family, at 72B and 235B.

SIX PROMPTING STRATEGIES

standardchain of thoughtIRACword-by-word check

concise instructionself-consistency vote

THREE SUBMITTED RUNS

RUN AND AGGREGATION	VAL / TEST
DU3 plain majority vote	91.9 / 93.9
DU2 vote, deliberation on ties	92.6 / 96.3
DU1 hierarchical meta-ensemble	93.0 / 96.3

Why diversity wins, on the validation set. Different families disagree on 17.5% of questions, against 11.4% within a family.

87.2

one model with itself

91.9

nine-expert vote

93.0

meta-ensemble

No single method wins everywhere. Different legal tasks reward different inductive biases.
Tasks 1 and 2 are scored by F1, Tasks 3 and 4 by accuracy. The Pilot reports tort accuracy and rationale F1.

Task 1
Legal Case Retrieval

0.314
F1 · 11th of 54 runs

Post-submission: F1 0.346 with a deeper reranker

Method. Learning to rank with a gradient-boosted model over 34 hand-built features: 11 retrieval, 16 structural, 3 authority, and 4 temporal.

A merged BM25 and dense pool holds the correct case 99.1% of the time. The date and self-match filter adds 2.7 points.

Ablation, development set. F1 rises from 0.165 with BM25 alone to 0.312 with all 34 features, and 0.357 with the full training set.

The strongest single feature is the reciprocal-rank fusion of the two retrievers. The submitted run scores precision 0.295 and recall 0.337.

Reason. Query documents have a median length of 4,573 tokens, beyond the 512-token window of the neural rerankers, which truncate and fail.

Finding. On long documents, hand-built features beat neural matching.

Task 2
Legal Case Entailment

0.343
F1 · 22nd of 35 runs

Post-submission: F1 0.555, above the best official entry of 0.490

Method. BM25 returns 100 candidate paragraphs, a MonoT5 and a Qwen3 reranker cut them to 20, then three open-weight models vote on entailment.

Cases hold 2.94 entailing paragraphs on average, so a single-selection rule caps F1 at 0.508. Of 230 gold paragraphs the official run missed, 122 were lost to that cap alone.

Multi-selection. The three models reach F1 0.549, 0.533, and 0.501. Their vote reaches 0.555, above any one of them.

Reason. The official prompt asked for one paragraph. Changing it to select every entailing paragraph lifted F1 to 0.555.

Finding. The wording of the instruction mattered more than the choice of model.

Task 3
Statute Retrieval and Entailment

79.3%
accuracy · 65 of 82 · 14th of 22

Post-submission: 91.5% with Qwen3-235B and an IRAC prompt

Method. Character-bigram BM25 with cross-reference expansion returns 11 to 17 articles per question, then Qwen2.5-72B makes a structured entailment decision at temperature zero.

The 82 questions come from the Japanese bar examination, each answered Yes or No against the 725-article Civil Code.

Retrieval recall is between 0.94 and 0.98, so the entailment model is the limit. A more elaborate run lost 3.7 points to the simpler one.

Entailment model. Qwen2.5-72B gives 79.3%, the nine-expert ensemble 86.6%, and Qwen3-235B with IRAC 91.5%.

Reason. With retrieval near saturated, a stronger entailment model gave the gain, not more pipeline.

Finding. Added complexity hurt. The ensemble that won Task 4 is misled here by distractor articles.

Pilot
Legal Judgment Prediction

73.1% 68.2%
tort accuracy · 587 of 803 rationale F1 · all

Unofficial (wrong run mode), yet above every official entry

Method. Five views read each case: two joint BERT models, a claim-only BERT, a retrieval view, and an entailment model. A stacker weights them, a claim-to-verdict bridge repairs the decision, and a coherence check removes contradictions.

Five views → Stacker weights → Claim-to-verdict bridge → Coherence check

The retrieval view earns the largest stacker weight, 1.86, the entailment model the smallest, 0.11. The bridge recovers material lost when 41% of long cases are truncated at 512 tokens, worth 2.8 points.

Ablation. A single BERT reaches 70.9%, the five views without the bridge 70.5%, the full system 73.1%.

Finding. The gain comes from the verdict bridge. Without it the five views do no better than a single model.

Cross-task takeaways

No method wins everywhere.
Each task asks for a different bias.

Diversity beats scale.
Task 4. Disagreement across families wins.

Restraint beats complexity.
Task 3. The simpler decision held up.

The prompt can outweigh the model.
Task 2. Wording set the ceiling.

What did not work on Task 4. Five learned combiners scored between 85.3 and 90.3%, below the plain vote. The best fine-tuning, ELYZA-8B with a low-rank adapter, reached 77.3%. Diversity in the prompt and the model did more for us than training on the task.

First place at COLIEE 2026 Task 4, 96.3%.
Department of Computer Science, Durham University, United Kingdom
COLIEE 2026 at ICAIL 2026. Singapore Management University, 8 to 12 June 2026.

Every system uses open-weight models only. For Tasks 3 and 4, all models were released before 15 July 2025, as the rules require.

In the repository

Full code and prompts for all five tasks.

- The exact prompts used for entailment
- The 34 retrieval features and the rerankers
- The pilot solver and the nine-expert vote
- The evaluation scripts and the per-task results

Open-weight models only. Released under the MIT licence.



Scan to open the repository
github.com/alshehriamal1994/team-du-coliee-2026